

Projects Deep-Dive & Interview Defense

How each project works, what every number means, and how to answer when grilled

Prepared for Saikumar Kaleru. If an interviewer says "walk me through this project," everything you need is here. Read it until you can explain each project without notes.

Project numbers below match your project folder: `1_nifty_banknifty_backtester`, `2_daily_technical_research_report`, `3_technical_coverage_watchlist`.

The portfolio narrative (say this if asked "tell me about your projects")

"I built three projects that mirror the technical-research workflow end to end: a **backtester** that validates whether technical strategies actually work, a **research-report generator** that produces the daily/weekly/monthly notes an analyst publishes, and a **coverage watchlist** that screens a sector universe for setups. The reports and watchlist are the analyst's daily output; the backtester is where my coding background adds an edge — I can test an idea on data instead of trusting it blindly."

Golden rule for all three: they use **free market data (yfinance)**, and I **coded the indicators by hand** (moving averages, RSI, MACD, ATR, Bollinger, candlestick rules) — so I understand the math, I'm not just calling a black box.

PROJECT 1 — Multi-Strategy Technical Backtester

Tool: Python. This is my differentiator — coding lets me test an idea on data instead of trusting it blindly.

The one-line pitch (memorise this)

"I built a backtesting engine in Python that tests eight technical strategies on Nifty 50 and Bank Nifty over ~6 years of daily data and compares each one against buy-and-hold on risk-adjusted metrics like CAGR, Sharpe and max drawdown — so a strategy is validated with evidence before anyone trades it."

Why this matters for a TRA

A research analyst recommends trade ideas. **Backtesting is how you prove an idea has an edge instead of guessing** — it's the difference between "RSI looks good" and "RSI mean-reversion returned X% with a Sharpe of Y over six years." It shows I can quantify whether a strategy actually works.

The eight strategies (be able to describe each)

All are **long/flat** — you either own the index (position = 1) or sit in cash (position = 0). No short-selling, no leverage.

#	Strategy	Type	Buy signal	Exit signal
1	MA Crossover + RSI	trend	20-MA > 50-MA and RSI > 50	when that's no longer true
2	RSI Mean-Reversion	mean-revert	RSI < 30 (oversold)	RSI > 55 (bounce done)
3	MACD Trend	trend	MACD line > signal line	MACD line < signal line
4	Bollinger Breakout	breakout	close breaks above upper band	close falls below the 20-MA
5	Golden Cross 50/200	trend	50-MA > 200-MA (golden cross)	50-MA < 200-MA (death cross)
6	Bollinger Mean-Reversion	mean-revert	close dips below the lower band	close back above the 20-MA
7	Stochastic Reversal	mean-revert	%K < 20 and crossing up through %D	%K > 80
8	Donchian Breakout	breakout	close > prior 20-day high (Turtle)	close < prior 10-day low

Simple on/off rules (MA-cross, MACD, golden cross) are one-line conditions. Mean-reversion and breakout strategies are **stateful** — a helper "remembers" you're in the trade and keeps you in until the exit fires. (*Why? RSI < 30 triggers the buy, but you stay in while RSI climbs to 55 — a plain condition would drop you out the very next day.*)

The result worth quoting: on Nifty, **Bollinger Mean-Reversion matched buy-and-hold's Sharpe (1.16) with half the drawdown** (−10.4% vs −17.2%) and an 88% win-rate, while invested only ~16% of the time — a clean example of a *risk-adjusted* edge even when raw return trails a bull market.

How it works — the 5-step pipeline (data → indicators → positions → performance → report)

1. **Get the data.** Download 6 years of daily Open/High/Low/Close for Nifty (`^NSEI`) and Bank Nifty (`^NSEBANK`) via the free **yfinance** library — about 1,500 trading days each.
2. **Compute indicators by hand (pandas).** I calculate them myself, not via a black-box library, so I can explain the formula: **SMA20/SMA50** (`.rolling().mean()`), **RSI(14)** = $100 - 100 / (1 + \text{avgGain}/\text{avgLoss})$, **MACD** = 12-EMA − 26-EMA with a 9-EMA signal line, and **Bollinger Bands** = SMA20 ± 2 standard deviations.
3. **Turn indicators into a position.** Each strategy outputs the 0/1 position series above.
4. **The single most important line — no look-ahead bias.** The position is **shifted by one day**: today's signal is acted on *tomorrow*. You can't trade on a close you haven't seen yet — shifting the position by one bar removes look-ahead bias and makes the backtest realistic. **If they ask one hard question about this project, it's this.**
5. **Measure & report.** Daily strategy return = (yesterday's position) × today's market return, compounded into an **equity curve**; individual **trades** (entry→exit) are extracted for win-rate and profit factor; the full metric set

is computed; and it renders an equity-curve chart plus a landscape **PDF** with a metrics table per index (the buy-and-hold row highlighted as the benchmark).

Every metric explained (they WILL ask)

Metric	Plain meaning
Total Return %	How much ₹1 grew over the whole 6 years
CAGR %	That return as a smooth <i>annual</i> growth rate
Sharpe	Return per unit of total volatility; higher is better, >1 is good
Sortino	Like Sharpe but counts only downside volatility (upside swings aren't "risk")
Max DD %	Worst peak-to-trough fall — the most pain you'd have sat through
Win %	Share of trades that were profitable
Profit Factor	Gross profit ÷ gross loss; above 1 means winners outweigh losers
Exposure %	Fraction of the time actually invested (the rest in cash — no return, no risk)

The honest, impressive talking point

"The active strategies **underperformed buy-and-hold** over this period — because it was a strong bull market and being in cash part of the time cost returns. But some had a **better Sharpe or a smaller max drawdown**, meaning they took less risk to get there. The real lesson was learning to judge a strategy honestly with Sharpe, Sortino and drawdown — *risk-adjusted* returns and the market regime — instead of chasing a back-fitted 'winner'."

(Admitting the strategies lost to buy-and-hold shows maturity and that you didn't fake results — this impresses far more than a suspiciously perfect backtest.)

Likely questions & answers

- **"How did you avoid look-ahead bias?"** → "I act on the *previous* day's signal — I shift the position by one bar, so I never trade on information I wouldn't have had in time."
- **"Did you calculate RSI/MACD yourself?"** → "Yes, by hand in pandas, so I understand the formulas rather than calling a black box."
- **"What's overfitting / curve-fitting?"** → "Tuning a strategy so perfectly to past data that it fails live. I used fixed, standard parameters instead of optimising them to flatter the result."
- **"Why long/flat, no shorting?"** → "Keeps it realistic for cash-index trading — you're either invested or in cash."
- **"Why Sortino as well as Sharpe?"** → "Sharpe penalises *all* volatility, including good upside moves; Sortino only penalises downside, which is what actually hurts."
- **"Why did buy-and-hold win?"** → "Strong uptrend + the active strategies sit in cash part of the time (see the Exposure column). In a sideways or bear market their lower drawdown would matter more."
- **"What would you add?"** → "Transaction costs and slippage, out-of-sample / walk-forward testing, and position sizing — right now it's all-in / all-out."

PROJECT 2 — Daily / Weekly / Monthly Technical Research Reports

Tools: Excel + Python. This is the actual job.

What it does (one line)

Produces a research note for **Nifty 50, Bank Nifty, Gold and Crude Oil** on three timeframes, each giving a **Buy/Sell/Hold** call with entry, target and stop-loss, backed by indicators and a chart.

How it works — step by step

1. **Pull data** for each asset (yfinance). For weekly/monthly, the daily data is **resampled** into weekly (Friday close) or monthly (month-end) candles using Open=first, High=max, Low=min, Close=last.
2. **Compute indicators** on the last bar: 20- and 50-period moving averages, RSI(14), MACD(12/26/9), Bollinger Bands (20, $\pm 2\sigma$), and ATR(14).
3. **Find levels**: support and resistance from the recent swing low/high, and Fibonacci retracement levels (23.6 / 38.2 / 50 / 61.8%) of that swing.
4. **Score the view**: a points system — +1 if price > 20-MA, +1 if 20-MA > 50-MA, +1 if MACD > signal, ± 1 for RSI extremes, ± 1 for Bollinger position. Score $\geq +2$ = BUY, ≤ -2 = SELL, else HOLD.
5. **Guardrail**: if the model says SELL but RSI < 25 (deeply oversold), it steps aside to HOLD — you don't short a tape that's already collapsed. Same for chasing an overbought BUY (RSI > 80).
6. **Set risk levels**: stop-loss = $1.5 \times \text{ATR}$ from entry, target = $2.5 \times \text{ATR}$ — so the levels are sized to each asset's actual volatility ($\approx 1:1.7$ reward-to-risk).
7. **Render** an annotated chart + a styled PDF per timeframe.

What the output shows

A clean PDF per timeframe with one card per asset: the call badge (green/red/amber), entry/target/stop, a table of all indicator values, Fibonacci levels, and a 6-month/2-year chart with support, resistance and Fib lines drawn.

The real artifacts (what to actually open in the interview)

This is genuinely an **Excel + Python** workflow: - **Daily_Levels_Tracker.xlsx** — an Excel workbook I maintain each morning. The **Pivot Levels** sheet takes yesterday's High/Low/Close and computes pivot, R1/R2 and S1/S2 with live formulas ($P = (H+L+C)/3$, $R1 = 2P - Low$, $S1 = 2P - High$); the **Trade Calls** sheet auto-calculates reward:risk for each entry/stop/target. - **Python** — automates the same analysis across four assets and three timeframes so it stays current.

Demo line: "I keep the levels and calls in this Excel workbook — here's the formulas — and I automated the repetitive part in Python so the daily note builds itself."

The standout talking point (multi-timeframe)

"Gold recently flagged **SELL on the daily** but **BUY on the monthly** — that's a short-term pullback inside a long-term uptrend. A good analyst reconciles timeframes: I'd be cautious shorting it because the bigger trend is up."

Likely questions & answers

- **"Why ATR-based stops instead of a fixed %?"** → "A fixed 2% stop is too tight for Bank Nifty and too wide for a calm stock. ATR sizes the stop to how much the asset actually moves, so I'm not stopped out by normal noise."
- **"What if all your indicators disagree?"** → "That's a HOLD. I only act on confluence; mixed signals mean no edge."
- **"Is this automated or your analysis?"** → "The data and levels are automated in Python to save time; the judgement — which call to publish and how to frame it — is mine. Automation just keeps it current."

PROJECT 3 — Multi-Asset Technical Coverage Watchlist

Tools: Excel + Python.

What it does (one line)

Screens **21 assets across 6 sectors** (indices, banking, IT, energy/materials, auto/FMCG, commodities) and tags each **Buy/Sell/Hold** with a ★ strength rating and a stop-loss — a morning coverage sheet for relationship managers.

How it works — step by step

1. Pull 2 years of daily data for each name.
2. Compute the **daily trend** (from 20/50-MA and price) and, by resampling, the **weekly trend** — this is the **multi-timeframe confluence**.
3. Compute RSI, MACD, a candlestick read (engulfing/hammer/doji/shooting-star), ATR.
4. **Composite score** across daily trend + weekly trend + MACD + price-vs-50MA + RSI extremes + candlestick.
The ★ **strength** reflects how many factors align.
5. Apply the same oversold/overbought guardrail; set an ATR stop on actionable calls.
6. Group by sector, sort by conviction, and render a one-page PDF + CSV with a Buy/Sell/Hold summary count.

The real artifacts (what to actually open in the interview)

An **Excel + Python** coverage workflow: - **Coverage_Watchlist.xlsx** — 21 assets grouped by sector. I read daily trend, weekly trend, RSI and MACD and drop them in; the **View** column computes via formula (daily Up *and* weekly Up = BUY; both Down = SELL; else HOLD), the ★ **Strength** counts how many factors align, RSI cells colour red/green at 70/30, and a **Summary** box counts BUY/SELL/HOLD with **COUNTIF** . - **Python** — runs the same scan automatically across all 21 names.

Demo line: "This is my Excel coverage sheet — it scores each name from the reads I enter; the Python version just does the whole scan for me in one run."

The edge — multi-timeframe confluence

"The strongest setups are where the **daily and weekly trends agree**. A name that's up on both is a higher-conviction buy than one that's up on the daily but down on the weekly — that second one is just a bounce in a downtrend."

Likely questions & answers

- "How do you rank what to look at first?" → "By conviction — the ★ strength — within each sector, so an RM can scan the strongest setups per sector instantly."
 - "Why group by sector?" → "Sector rotation matters; an RM covering banking wants the banking view together, and clustering shows when a whole sector is trending."
 - "Isn't this just the report for more stocks?" → "Different purpose — the report is a deep note on a few assets; the watchlist is broad coverage to *spot* where to look. They're the scan and the deep-dive."
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HONEST LIMITATIONS (acknowledging these shows maturity)

If asked "what are the weaknesses?", say any of these — it signals you think critically: - **No transaction costs / slippage** in the backtest yet — real returns would be a bit lower. - **Rules-based, not discretionary** — the scoring is mechanical; a human analyst adds context (news, events, macro) the model can't see. - **Commodity data uses global benchmarks** (COMEX gold, WTI crude) as proxies for MCX — the technicals transfer, but exact MCX levels differ with USD/INR and duties. - **Backtest is in-sample** — I'd add walk-forward / out-of-sample testing to be rigorous.

60-SECOND ELEVATOR VERSION (if they want it short)

"Three projects covering the analyst workflow: a **backtester** that tested eight strategies over ~6 years on proper metrics — Sharpe, Sortino, drawdown, profit factor; a Python-automated **research report** for indices and commodities across daily/weekly/monthly with Buy/Sell/Hold calls and ATR stops; and a **sector watchlist** that screens 21 assets on daily+weekly confluence. The honest finding from the backtest was that the simple strategies didn't beat buy-and-hold in a bull market, which taught me to validate ideas with data rather than trust them."

Run everything before the interview with: `python run_all.py` — outputs land in each project's `output/` folder. Pair this with the Handbook (theory) and the crash course (rapid revision).